# Project Report

A study of RAG technique for generating image captions

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# *Abstract*

Automatic image caption generation (image captioning) is a fundamental task in computer vision and natural language processing. Retrieval-Augmented Generation (RAG) techniques enable the generation process to be grounded in external knowledge (similar images, reference captions, concept graphs) in order to improve the accuracy, relevance and visual grounding of the generated captions.

This project explores the integration of Retrieval-Augmented Generation (RAG) with image captioning, focusing on the **Self-Reflective** Retrieval-Augmented Generation (Self-RAG) technique. The proposed approach combines visual information extracted from input images with relevant textual descriptions retrieved from an external knowledge base constructed using a subset of the *MS COCO* dataset.

Ultimately, the system aims to evaluate the relevance of retrieved information and improve the reliability of generated captions. The project also investigates the effectiveness of this approach through quantitative and qualitative evaluation, with particular attention to caption quality, contextual relevance, and hallucination reduction.

**Keywords:** Image captioning, RAG, self-RAG, Computer vision, MS COCO, Hallucination.

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# I. Presentation Of the Project Context

## A. Introduction

The rapid development of Artificial Intelligence, particularly in the fields of Computer Vision and Natural Language Processing (NLP), has significantly improved the ability of machines to understand and interpret visual information. Among the various applications emerging from these fields, image analysis and image captioning have become an important research area. It aims to enable computers to automatically generate natural language descriptions of images by identifying objects, understanding their relationships, and interpreting the overall scene.

Image captioning represents a challenging task because it requires the integration of two distinct capabilities**: visual understanding** and **language generation**. A system must first analyze an image, recognize its relevant visual elements, and extract meaningful information. It must then transform this information into a coherent, grammatically correct, and semantically accurate sentence.

Traditional image captioning approaches generally rely on deep learning architectures that combine visual feature extraction with language generation. Although these approaches have achieved considerable progress, they may still produce incomplete descriptions, overlook important visual details, or generate information that is not actually present in the image.

To address some of these limitations, Retrieval-Augmented Generation (RAG) has emerged as a promising approach. Rather than relying exclusively on the knowledge encoded within a model, RAG introduces an external knowledge source from which relevant information can be retrieved and incorporated into the generation process. This additional information can provide useful contextual references and support the production of more informative outputs.

Within this context, Self-Reflective Retrieval-Augmented Generation (Self-RAG) introduces an additional mechanism based on self-reflection. Instead of retrieving information indiscriminately, the model learns to determine when retrieval is necessary and to evaluate the relevance of retrieved information as well as the quality and factual support of its generated responses.

This project explores the application of Retrieval-Augmented Generation to image captioning, with a particular focus on the Self-RAG technique. The objective is to investigate how retrieval and self-reflection mechanisms can contribute to generating more accurate, relevant, and informative descriptions of visual content.

## B. Project Overview

The project focuses on the development and evaluation of a Retrieval Augmented Image Captioning system based on the principles of Self-RAG.

The proposed approach combines information extracted from an input image with relevant textual information retrieved from an external knowledge base. This knowledge base is constructed using image-caption pairs from the Microsoft Common Objects in Context (*MS COCO*) dataset.

When an image is provided to the system, its visual content is analyzed to identify relevant information. Based on this information, the system can retrieve descriptions associated with visually or semantically related examples from the knowledge base. These retrieved descriptions serve as additional contextual information during the caption generation process.

# II. Problem Statement

## A. Image Captioning Challenges

Image captioning is a complex task that consists of automatically generating a natural language description of the visual content of an image. Unlike image classification, which generally assigns one or several predefined labels to an image, image captioning requires a system to produce a meaningful sentence that describes the scene and its relevant elements.

To generate an accurate caption, a model must perform several interconnected operations. It must identify the objects present in the image, recognize their attributes, understand their spatial and semantic relationships, and determine which information is sufficiently important to be included in the final description.

For example, when analyzing an image containing a person riding a bicycle on a street, a captioning system should identify the person, the bicycle, and the action being performed. It should then combine these elements into a coherent sentence that accurately represents the scene.

However, several challenges make this process difficult.

- **Object recognition and scene understanding**
- **Relationship identification**
- **Contextual understanding**
- **Language generation**
- **Factual consistency**

These challenges demonstrate that successful image captioning requires more than object recognition. It requires a combination of visual understanding, contextual reasoning, and reliable language generation.

## B. Limitations of Traditional Image Captioning

Traditional image captioning systems commonly follow an encoder-decoder architecture.

The encoder processes the input image and extracts a visual representation, while the decoder uses this representation to generate a natural language description.

Early approaches relied on Convolutional Neural Networks (CNNs) for visual feature extraction and Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory networks (LSTMs), for sentence generation.

Although these approaches established important foundations for automatic image captioning, they present several limitations amongst them:

- Dependence on learned visual representaion.
- Limited access to external information.
- Incomplete scene descriptions.
- Hallucinations.
- Lack of explicit self-evaluation.

The objective is not simply to produce longer or more detailed descriptions. Instead, the system should generate captions that remain faithful to the input image while benefiting from relevant contextual information.

The project therefore investigates “How can RAG, combined with self-reflection mechanisms, improve the accuracy, relevance, and factual consistency of automatically generated image captions?”

# III. State of the Art: RAG and Image Captioning

## A. Introduction to RAG:

Retrieval-Augmented Generation (RAG) is an approach that combines information retrieval with natural language generation. It was developed to address limitations associated with relying exclusively on the knowledge stored within the parameters of a generative model. In a conventional generative system, the model produces an output based primarily on its learned parameters and the information provided in the input prompt.

In contrast, a RAG system introduces an external knowledge source. Before or during generation, the system retrieves information relevant to the current input and provides it as additional context. The general RAG process consists of three main stages:

**1. Retrieval**

The system receives an input query and searches an external knowledge base for relevant information.

**2. Augmentation**

The retrieved information is combined with the original input to provide additional context to the generative model.

**3. Generation**

The model uses the original input and the retrieved information to produce the final response.

However, traditional RAG systems also present limitations.

In particular, retrieval is often performed using a fixed strategy, regardless of whether additional information is necessary. Furthermore, retrieved documents may not always be relevant to the input, and the generative model may produce statements that are not adequately supported by the retrieved information.

These limitations have motivated the development of more adaptive and reflective retrieval approaches, including Self-RAG.

## B. Introduction to Self-RAG:

The framework aims to improve the quality and factual consistency of large language model outputs by combining retrieval with a self-reflection mechanism.

The main motivation behind Self-RAG is that conventional RAG systems retrieve a predetermined number of documents regardless of whether retrieval is actually necessary or whether the retrieved information is useful.

### I. Comparison Between Conventional RAG and Self-RAG

Although conventional RAG and Self-RAG share the same fundamental idea of combining retrieval with generation, they differ in how retrieval and evaluation are handled.

| **Aspect** | **Conventional RAG** | **Self-RAG** |
|---|---|---|
| Retrieval strategy | Follows generally a predefined retrieval procedure | Adaptively determines whether retrieval is necessary |
| Retrieval frequency | Often fixed | Vary according to the input and generation process |
| Relevance evaluation | May rely mainly on the retriever | Includes an explicit self-reflection mechanism |
| Generation | Conditioned on retrieved information | Conditioned on retrieved information and reflection mechanisms |
| Output evaluation | Often performed separately | Includes evaluation during generation |
| Main objective | Enrich generation with external information | Improve generation through adaptive retrieval and self-critique |

The main contribution of Self-RAG is simply not the addition of an external knowledge base. Its distinguishing feature is the integration of mechanisms that allow the model to assess the usefulness of retrieval and the quality of generated content.

This distinction is particularly relevant to image captioning, where irrelevant retrieved descriptions can introduce incorrect objects or relationships into the final caption.

# IV. System Architecture

## A. Overall Architecture:

The system receives an input image and processes its visual content to obtain a meaningful representation. This representation is then used to determine whether additional contextual information is required from an external knowledge base. When retrieval is necessary, the system searches for relevant descriptions associated with similar images and evaluates their relevance before incorporating them into the generation process.

Unlike conventional image captioning architectures, which generally generate descriptions directly from extracted visual features, the proposed architecture introduces an additional retrieval component and a self-reflection mechanism.

The project is structured around four main aspects:

- **Dataset preparation:** Selecting, organizing, and preprocessing image-caption pairs from the MS COCO dataset.
- **Knowledge base construction:** Building a collection of textual descriptions that can be used during retrieval.
- **Retrieval-Augmented Generation:** Integrating relevant retrieved information into the image captioning process.
- **Self-reflection:** Applying the principles of Self-RAG to evaluate retrieval relevance and the reliability of generated captions.

These components interact through a sequential but adaptive workflow. The system can decide whether external information is necessary, assess the quality of retrieved descriptions, and evaluate the generated caption before producing the final output.

# V. Dataset and Knowledge Base

## A. Images and Annotations Preprocessing :

Before constructing the knowledge base, the MS COCO 2017 dataset undergoes a preprocessing stage to ensure the quality, consistency, and proper organization of both images and their corresponding annotations. This stage is essential because the retrieval process depends on reliable image-caption associations and properly prepared data.

### 1. Image Preprocessing

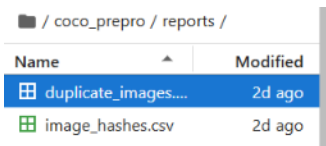
The first step consists of verifying the **integrity** of the downloaded images. Each image is checked to ensure that it can be successfully opened and processed. This operation helps identify corrupted or unreadable files that could cause errors during subsequent feature extraction and retrieval operations.

A preliminary integrity check was performed on the downloaded COCO 2017 image collection, containing 118,287 images. The results showed that all images were valid, with no corrupted files detected.

```
========== IMAGE INTEGRITY ==========
Total images checked: 118287
Corrupted images: 0
Valid images: 118287
```

-figure 1- presenting the result of integrity check script

Duplicate detection was also performed to identify repeated images within the collection. A total of 14 duplicate copies were detected, leaving 118,273 unique images. Removing these duplicates helps avoid unnecessary repetition of visual information within the dataset.




Delimiter: ,

| | image_name | sha256 | status | duplicate_of |
|---|---|---|---|---|
| 1 | 000000090169.jpg | d53a98667a2337908f9db | duplicate | 000000058627.jpg |
| 2 | 000000242807.jpg | b2f1beaf7ee77ef4d4bc9b | duplicate | 000000137715.jpg |
| 3 | 000000279522.jpg | 59d65728ec29e2e52a9a2 | duplicate | 000000019665.jpg |
| 4 | 000000297126.jpg | bd623c617f99ea0447381 | duplicate | 000000186275.jpg |
| 5 | 000000321238.jpg | 663fbba0f7fdca444decc1 | duplicate | 000000089027.jpg |
| 6 | 000000341448.jpg | 71da405bb842012a00588 | duplicate | 000000000927.jpg |
| 7 | 000000373199.jpg | 3cc51733e9c1bcdbda659 | duplicate | 000000308608.jpg |
| 8 | 000000397819.jpg | 716fc9dd6e936787faf181 | duplicate | 000000388662.jpg |
| 9 | 000000407259.jpg | 9ea9c483e77ecdfd3d722 | duplicate | 000000278029.jpg |
| 10 | 000000410810.jpg | df5a360e5533352a04c61 | duplicate | 000000373251.jpg |
| 11 | 000000411274.jpg | 16c4ae722b9ea38a4d6d2 | duplicate | 000000216265.jpg |
| 12 | 000000446141.jpg | 9ea9c483e77ecdfd3d722 | duplicate | 000000278029.jpg |
| 13 | 000000531721.jpg | cd43cf472fd4b0269445a5 | duplicate | 000000438867.jpg |
| 14 | 000000578492.jpg | 716fc9dd6e936787faf181 | duplicate | 000000388662.jpg |

-figure 2- the .csv report file containing all 14th detected duplicate

Following these verification operations, the selected images are prepared for further processing. Depending on the requirements of the visual encoder, this may include resizing, color format conversion, and pixel normalization to ensure consistency across the dataset.

### 2. Annotations Preprocessing

In addition to the images, the textual annotations provided by COCO 2017 require proper extraction and organization. The dataset provides five human-written captions for each image, offering different descriptions of the same visual scene.

The annotation files, provided in JSON format, contain information such as image identifiers, file names, and caption texts. These annotations are extracted and associated with their corresponding images using their unique image IDs.
The preprocessing of annotations involves extracting the five captions associated with each selected image, checking for missing or empty descriptions, and organizing the resulting information into structured image-caption pairs.

The original meaning of the captions is preserved to maintain the quality of the human-provided descriptions. Each image therefore remains linked to its corresponding textual annotations, ensuring that the retrieval module can access the correct descriptions when searching the **knowledge base**.

### 3. Final constructed knowledge base

After preprocessing, the images and annotations are organized according to their respective roles in the project. The selected 10,000 training images and their 50,000 captions are used to construct the knowledge base, while the additional validation and testing images and their reference captions are stored separately for future testing and evaluation.

This separation is particularly important to prevent *data leakage*. The reference captions associated with validation and test images must not be included in the knowledge base or made available to the retrieval mechanism during evaluation.

Overall, these preprocessing operations provide a clean and structured collection of image-caption pairs, establishing a reliable foundation for knowledge-base construction and the subsequent Self-RAG image captioning process.